

Unbreakable Steganography with LLMs

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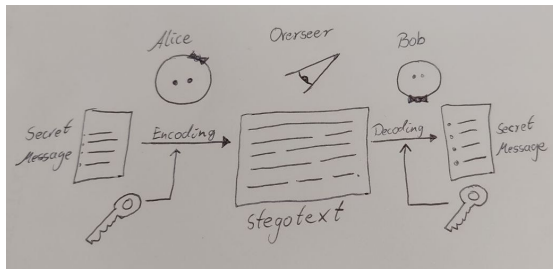
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1 Overview

2 Main Content

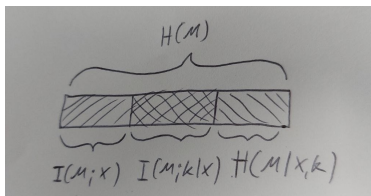
Steganography Setup



Ingredients:

- Sender/Encoder E and Receiver/Decoder D
- Communication Channel C
- A secret key k shared by E and D
- A distribution $\mathcal{D}$ of innocent messages
- A message space M

Can we have perfect perfect information theoretic security?



Perfect Steganography:

- M: Message, X: Stego-Text, K: Key
- Perfect Security: $I(M; X) = 0$
- Perfect Decoding: $H(M|X, K) = 0$

$$I(M; X, K) = H(M) - H(M|X, K) \quad (\text{Def of Mut. Info}) \quad (1)$$

$$I(M; X, K) = I(M; X) + I(M; K|X) \quad (\text{Chain Rule}) \quad (2)$$

$$\Rightarrow H(M) = \underbrace{H(M|X, K)}_{=0} + \underbrace{I(M; X)}_{=0} + I(M; K|X) \leq H(K) \quad (3)$$

$$\Rightarrow H(M) \leq H(K) \quad (4)$$

Computationally Secure Steganography

- Basic Idea: Security against polynomially bounded adversaries
- Uses Pseudo-Random Functions (PRF), i.e., deterministic functions

$$F = \left\{ F_k : \{0, 1\}^{|x|} \rightarrow \{0, 1\}^{|y|} \right\}_{k \in \{0, 1\}^{|k|}},$$

[$|k|, |x| = \text{poly}(|k|), |y| = \text{poly}(|k|)$ are key, input and output length]

- where, for any poly-time distinguisher D , the advantage

$$\text{Adv}_{F,D}^{\text{PRF}}(|k|) := \left| \Pr_k \left[D^{F_k}(1^{|k|}) = 1 \right] - \Pr_f \left[D^f(1^{|k|}) = 1 \right] \right|,$$

where

$$k \sim \mathcal{U} \left(\{0, 1\}^{|k|} \right) \quad \text{and} \quad f \sim \mathcal{U} \left(\{0, 1\}^{|x|} \rightarrow \{0, 1\}^{|y|} \right)$$

is negligible, i.e.,

$$\text{Adv}_{F,D}^{\text{PRF}}(|k|) < \frac{1}{\text{poly}(k)}$$

PRF + LLM = Unbreakable Steganography

- Any signal that has entropy (or random degrees of freedom), can be turned into an unbreakable stego scheme via PRFs
- Basic Idea:
 - Sender and Receiver share PRF scheme and secret key k
 - Sender uses LLM to generate distribution over next token
 - The PRF with the next symbol as input generates a pseudo-random sample
 - The receiver eliminates every symbol that wasn't compatible with the tokens so far until only one symbol remains
 - Sender moves on to next symbol

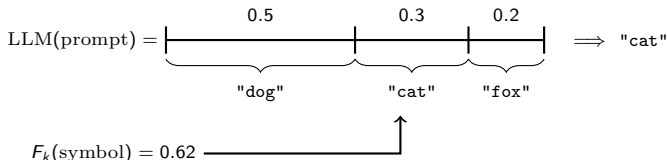


Figure: Using a PRF to select from the next-token distribution

- **Scenario 1:** LLM and prompt are known $\rightarrow$. Receiver can decode perfectly
- **Scenario 2:** Only tokenizer is known: Receiver can still decode probabilistically with error correction ($\leftarrow$ symbol)
- Receiver aggregates score for every symbol
- Once score exceeds threshold t , symbol is accepted
- If a wrong symbol got accepted, the sender sends a $\leftarrow$ correction symbol
- As long as the error rate is < 0.5 , the process has net progress towards correct symbols

Questions?